

K: Sample Median Preservation

Source file name: `samplemedian.c`, `samplemedian.cpp`, `samplemedian.java`, or `samplemedian.py`

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The National Metrology and Statistics Institute is conducting a quality control study on industrial sensor readings. They have collected a dataset consisting of N integer measurements $A_1, A_2, \dots, A_N$, where N is guaranteed to be an **odd** integer.

The true representative benchmark of this dataset is defined as its **median**: when all N measurements are sorted in non-decreasing order, the median is the unique value located exactly at index $(N + 1)/2$.

Due to computational constraints, the analysis team cannot always process the entire dataset. Instead, they choose a random sample of K measurements uniformly at random **without replacement**, where K is also an **odd** integer ($1 \leq K \leq N$). That is, each of the $\binom{N}{K}$ possible subsets of K distinct elements is equally likely to be selected. The sample median is similarly defined as the value at position $(K + 1)/2$ once the K chosen elements are sorted.

The team wants to evaluate the reliability of this sampling technique. What is the probability that the sample median equals the true median of the complete dataset?

Input

The input consists of several test cases.

Each test case begins with a line containing two odd integers N and K ($1 \leq K \leq N \leq 10^6$), the size of the dataset and the size of the sample, respectively.

The next line contains N integers $A_1, A_2, \dots, A_N$ ($-10^9 \leq A_i \leq 10^9$), the measurements in the dataset. Note that values may appear more than once.

The input terminates with a line containing:

0 0

which should not be processed.

It is guaranteed that the sum of N over all test cases in the input does not exceed 10^6 .

The input must be read from standard input.

Output

For each test case, print a single line with an integer representing the probability that the sample median equals the median of the complete dataset.

It can be proven that this probability can be expressed as an irreducible fraction P/Q , where $\gcd(Q, 10^9 + 7) = 1$. Output the integer value:

$$P \cdot Q^{-1} \pmod{10^9 + 7}$$

where Q^{-1} denotes the modular multiplicative inverse of Q modulo $10^9 + 7$, that is, the unique integer $0 \leq Q^{-1} < 10^9 + 7$ satisfying $Q \cdot Q^{-1} \equiv 1 \pmod{10^9 + 7}$.

The output must be written to standard output.

Sample Input	Sample Output
5 3	8000000006
1 2 3 4 5	1
5 3	1
1 2 2 2 3	2000000002
1 1	
42	
5 1	
10 20 20 20 30	
0 0	

Explanation of the sample cases:

- In the first test case ($N = 5, K = 3$, dataset $[1, 2, 3, 4, 5]$), the true dataset median is 3. Out of the 10 possible samples of size 3, exactly 4 samples ($\{1, 3, 4\}$, $\{1, 3, 5\}$, $\{2, 3, 4\}$, and $\{2, 3, 5\}$) have a median of 3. Thus, the probability is $4/10 = 2/5$. Since $5^{-1} \equiv 400\,000\,003 \pmod{10^9 + 7}$, the required output is $2 \cdot 400\,000\,003 \equiv 800\,000\,006 \pmod{10^9 + 7}$.
- In the second test case ($N = 5, K = 3$, dataset $[1, 2, 2, 2, 3]$), the true dataset median is 2. Every sample of size 3 chosen from this dataset contains the median 2 at its center position, so the sample median is always 2. The probability is $1 \equiv 1 \pmod{10^9 + 7}$.